

The Born Perceptron: Mimicking Quantum Mechanics and Distributed Measurements in Time

A. Chawla

REAL Institute and IIT Delhi

Email: editor@realinstitute.live

Abstract—This paper develops a neural architecture capable of reproducing several structural aspects of quantum mechanics. The proposed model, termed the Born Perceptron, extends the classical perceptron by structuring outputs as value–index pairs and interpreting normalized activations as probability distributions over indexed outcomes. By allowing indices to inhabit geometric manifolds rather than discrete labels, the perceptron naturally represents probabilistic sampling over state spaces. When stacked sequentially, such perceptrons generate stochastic trajectories reminiscent of quantum state evolution under repeated measurement. Introducing complex-valued activations yields probabilities determined by squared magnitudes, reproducing the Born rule. The framework is further extended to treat measurements distributed in time, particularly position samplings. In this regime, the neural system produces sequences of sampled positions whose joint probabilities resemble those derived in path-integral formulations of quantum mechanics. The resulting architecture offers a classical computational model that captures several conceptual features of quantum theory including measurement, collapse, trajectory evolution, and amplitude-based probability rules.

I. INTRODUCTION

Quantum mechanics is fundamentally probabilistic. The outcome of a measurement cannot in general be predicted deterministically from the quantum state. Instead, measurement results appear as random samples drawn from probability distributions determined by the Born rule. For a quantum state represented by a density matrix ρ and a measurement described by a positive operator-valued measure (POVM) E_k , the probability of observing outcome k is given by

$$p_k = \text{Tr}(E_k \rho). \quad (1)$$

This probabilistic structure bears a striking resemblance to the behavior of classical machine learning models that produce normalized probability distributions. Neural networks routinely map input vectors to collections of scores that are normalized into probabilities using functions such as softmax. While such models were not originally designed to mimic quantum measurement, their mathematical structure provides an intriguing analogy.

The present work explores whether a simple neural primitive, the perceptron, can be structured to emulate several aspects of quantum measurement and evolution. We introduce the concept of the Born Perceptron, a perceptron whose outputs are interpreted as probability distributions over indexed outcomes. By pairing outputs into value–index structures and allowing the indices to reside in geometric manifolds, the

model can represent distributions over state spaces. Sampling from these distributions corresponds to a form of collapse analogous to quantum measurement.

When such perceptron layers are stacked sequentially, the sampled outputs propagate through the network and generate stochastic trajectories across the manifold of states. This behavior parallels descriptions of quantum trajectories in open quantum systems. By further extending the perceptron to support complex-valued activations, the squared magnitude of the resulting amplitudes determines probabilities, reproducing the mathematical structure of the Born rule.

The framework is extended to address measurements distributed in time. In particular, we consider the problem of sequential position sampling of a quantum particle. Path-integral formulations of quantum mechanics demonstrate that joint probabilities for sequences of measurement outcomes can be expressed in terms of integrals over all possible system trajectories. We show that a neural architecture built from stacked Born perceptrons can generate analogous joint distributions through sequential sampling and propagation.

II. STRUCTURED OUTPUTS AND THE PAIRING PRINCIPLE

Traditional perceptrons compute a vector of activations

$$v_k = w_k \cdot x + b_k \quad (2)$$

for an input vector x . In standard classification tasks the predicted class corresponds to the index of the largest activation.

In the Born perceptron this interpretation is modified. Instead of treating outputs as mere scores associated with categorical labels, each activation is paired with an explicit index. The output of the perceptron therefore takes the form

$$(v_1, i_1), (v_2, i_2), \dots, (v_n, i_n). \quad (3)$$

The values v_k are normalized into probabilities

$$p_k = \frac{e^{v_k}}{\sum_j e^{v_j}}. \quad (4)$$

The perceptron thus defines a conditional probability distribution over indices

$$x \rightarrow p(i_k | x). \quad (5)$$

Sampling from this distribution produces a single index i_k . This process resembles measurement collapse in the sense that a probabilistic distribution is reduced to a single realized outcome.

The pairing principle emphasizes that the perceptron output contains two distinct components. The value determines the probability weight associated with an outcome, while the index specifies the identity of that outcome. By explicitly representing both components, the perceptron can be interpreted as a probabilistic generator of outcomes rather than merely a classifier.

III. INDICES AS ELEMENTS OF GEOMETRIC MANIFOLDS

In conventional neural classifiers indices correspond to discrete labels. The Born perceptron generalizes this notion by allowing indices to occupy positions in a geometric space. Instead of treating i_k as an abstract label, we interpret it as a vector

$$i_k \in \mathbb{R}^d. \quad (6)$$

The output structure therefore becomes

$$(v_k, i_k) \quad (7)$$

with probabilities

$$p(i_k|x). \quad (8)$$

Sampling selects a point from a discrete set embedded within a continuous manifold. The perceptron therefore defines a probability distribution over candidate points in that manifold.

A useful example arises in the representation of qubit states. A qubit can be described by a Bloch vector

$$\mathbf{r} = (r_x, r_y, r_z) \quad (9)$$

lying within the Bloch sphere. If candidate indices correspond to points inside this sphere, the perceptron can generate distributions over possible quantum states. Sampling then selects a particular state vector.

In this interpretation the perceptron maps an input vector into a distribution over geometric states. The sampling process transforms the input into another point in the manifold, which can subsequently be propagated through further neural layers.

IV. RELATIONSHIP WITH POVM MEASUREMENTS

Quantum measurement probabilities exhibit a linear dependence on the quantum state. If the density matrix is vectorized into a real parameter vector x , the probability rule can be written as

$$p_k = w_k \cdot x. \quad (10)$$

This expression has precisely the same structure as a perceptron activation. Consequently, a perceptron can approximate the mapping from quantum states to measurement probabilities by learning appropriate weights.

Consider a qubit measured along the z axis. The probabilities of obtaining spin up or spin down are

$$p_{\uparrow} = \frac{1 + r_z}{2}, \quad p_{\downarrow} = \frac{1 - r_z}{2}. \quad (11)$$

A perceptron with weights aligned along the z direction reproduces this mapping. The correspondence highlights how the perceptron naturally implements linear probability rules similar to those governing POVM measurements.

V. STACKING LAYERS AND STOCHASTIC TRAJECTORIES

A single Born perceptron layer generates a probability distribution over indices and produces a sampled outcome. When such layers are arranged sequentially, the sampled outcome from one layer becomes the input to the next. If x_t denotes the state produced by layer t , the next layer samples a new index according to

$$x_{t+1} \sim p(i|x_t). \quad (12)$$

The resulting evolution

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \quad (13)$$

defines a stochastic trajectory across the manifold of indices. The probability of observing a particular trajectory is given by

$$P(\text{path}) = \prod_t P(x_{t+1}|x_t). \quad (14)$$

This structure is mathematically equivalent to a Markov chain. In the context of quantum measurement it resembles quantum trajectory descriptions in which a quantum system evolves stochastically under repeated observation.

VI. COMPLEX ACTIVATIONS AND THE BORN RULE

A closer analogy with quantum mechanics emerges when perceptron activations are allowed to be complex numbers. Let

$$z_k = w_k \cdot x \quad (15)$$

with $z_k \in \mathbb{C}$.

Probabilities are then obtained through squared magnitudes

$$p_k = \frac{|z_k|^2}{\sum_j |z_j|^2}. \quad (16)$$

This expression is identical in form to the Born rule used in quantum mechanics. The activations represent probability amplitudes whose squared magnitudes determine outcome probabilities.

As a simple illustration, consider complex activations

$$z_1 = 1 + i, \quad z_2 = 2 - i. \quad (17)$$

Their squared magnitudes are

$$|z_1|^2 = 2, \quad |z_2|^2 = 5. \quad (18)$$

The resulting probabilities are

$$p_1 = \frac{2}{7}, \quad p_2 = \frac{5}{7}. \quad (19)$$

Complex activations therefore allow perceptrons to represent amplitude-based probability structures.

VII. MEASUREMENTS DISTRIBUTED IN TIME

Quantum measurements frequently occur not as isolated events but as sequences distributed in time. In path-integral formulations the joint probability of a sequence of measurement outcomes can be expressed in terms of integrals over all possible trajectories of the system.

Consider a particle whose position is sampled at times

$$t_1 < t_2 < \dots < t_Q. \quad (20)$$

The measurement results are positions

$$x_1, x_2, \dots, x_Q. \quad (21)$$

In path-integral formulations the joint amplitude associated with this sequence involves contributions from all possible paths consistent with the sampled positions.

A neural architecture built from Born perceptrons can emulate this situation by performing sequential sampling. At each time step the network generates a distribution over possible positions and samples one value. The sampled position then influences the next step of the evolution.

VIII. NEURAL POSITION SAMPLING

Let the state of the system at time t be represented by a neural vector s_t . A measurement layer computes activations corresponding to candidate positions

$$v_k = w_k \cdot s_t. \quad (22)$$

After normalization, these activations define a probability distribution over position indices x_k . Sampling yields a measured position x_q .

Following the measurement, the neural state is updated using a transformation that incorporates the measurement outcome. This transformation may be interpreted as the neural analogue of a resolution function that modifies the wavefunction following a measurement.

The updated state then propagates forward to the next time step, where the process repeats. In this way the network generates a sequence of sampled positions.

IX. NEURAL REPRESENTATION OF PATH INTEGRALS

The sequential sampling procedure described above produces trajectories

$$x_1, x_2, \dots, x_Q \quad (23)$$

with probabilities determined by the product of conditional distributions generated by the perceptron layers.

When activations are complex-valued, each transition carries a complex amplitude. The amplitude associated with an entire trajectory is obtained by multiplying the amplitudes of individual steps. Summing amplitudes across trajectories yields interference effects analogous to those found in quantum mechanics.

In the limit of finely spaced time steps, the neural architecture approximates a path-integral formulation in which trajectories contribute amplitudes determined by accumulated phase factors.

X. DISCUSSION

The Born perceptron provides a classical computational architecture that captures several conceptual elements of quantum mechanics. The pairing of values and indices transforms perceptron outputs into probabilistic generators of outcomes. Allowing indices to inhabit geometric manifolds enables the model to represent distributions over state spaces rather than discrete labels. Sequential stacking of perceptron layers generates stochastic trajectories across these manifolds, mimicking the evolution of quantum systems under repeated measurement.

Complex activations introduce amplitude-based probability rules identical in form to the Born rule. When measurements are distributed in time, the resulting neural architecture produces sequences of sampled positions whose joint probabilities resemble those derived from path-integral formulations.

While the framework does not claim to reproduce the full physical structure of quantum theory, it offers a compelling conceptual bridge between neural computation and quantum probability. The approach suggests that several mathematical features of quantum mechanics may be interpreted as emergent properties of probabilistic sampling processes implemented by layered computational systems.

XI. CONCLUSION

This paper introduced the Born perceptron as a neural architecture capable of mimicking several structural features of quantum mechanics. By structuring perceptron outputs as value-index pairs and interpreting normalized activations as probability distributions, the model generates stochastic measurement outcomes analogous to those described by the Born rule. Allowing indices to occupy geometric manifolds enables the network to represent distributions over state spaces.

Stacking layers of Born perceptrons produces stochastic trajectories reminiscent of quantum state evolution under repeated observation. The introduction of complex-valued activations yields probability amplitudes whose squared magnitudes determine outcome probabilities. Extending the framework to measurements distributed in time enables the neural architecture to produce sequences of position samplings whose joint distributions resemble those obtained in path-integral formulations.

These results suggest that neural architectures based on probabilistic sampling may provide useful conceptual tools for understanding quantum measurement and dynamics. Future work may explore more detailed connections between neural trajectory models and the mathematical structures underlying quantum theory.

XII. NEURAL MEASUREMENT ARCHITECTURES AND THE BORN RULE

A. Quantum Measurement as Probabilistic Sampling

Quantum mechanics differs fundamentally from classical mechanics in that it does not predict deterministic outcomes for individual measurements. Instead, the theory provides probability distributions over possible outcomes. For a system

described by a density matrix ρ and a measurement represented by a set of positive operators $\{E_k\}$ forming a positive operator-valued measure (POVM), the probability of observing outcome k is

$$p_k = \text{Tr}(E_k \rho). \quad (24)$$

The result of a measurement is therefore a random variable drawn from this distribution. Once an outcome is observed, the system state is updated according to the measurement postulate, a process often described as wavefunction collapse.

A sequence of measurements performed at successive times produces a stochastic sequence of outcomes. These outcomes can be interpreted as forming a trajectory through the space of possible measurement results. In modern quantum theory such trajectories are used to describe continuously monitored systems and form the basis of quantum trajectory formulations of measurement dynamics.

This probabilistic character suggests an intriguing connection with computational systems that naturally produce probability distributions. In particular, many neural network architectures generate normalized probability outputs that are subsequently sampled to produce discrete outcomes. The structural similarity between probabilistic neural inference and quantum measurement motivates the exploration of neural architectures capable of reproducing elements of quantum measurement theory.

B. Probabilistic Neural Generators

Neural networks routinely produce probability distributions as outputs. A simple example is the perceptron layer, which computes a set of activations

$$v_k = w_k \cdot x + b_k \quad (25)$$

for an input vector x . These activations are typically converted into probabilities using a normalization such as the softmax function

$$p_k = \frac{e^{v_k}}{\sum_j e^{v_j}}. \quad (26)$$

Sampling from this distribution yields a discrete outcome k .

In conventional machine learning tasks, the index k corresponds to a class label. However, the perceptron can also be interpreted more generally as a probabilistic generator of outcomes. Each output unit produces a pair consisting of an activation value and an associated index,

$$(v_k, i_k). \quad (27)$$

The value determines the probability weight, while the index identifies the outcome. Sampling selects one of the indices according to the computed probabilities. In this interpretation the perceptron acts not merely as a classifier but as a stochastic generator of discrete events.

This perspective is particularly useful when modeling processes in which outcomes correspond to physical states rather than abstract labels.

C. Quantum States and Measurement Operators

The probabilistic nature of quantum measurement arises from the structure of quantum states and measurement operators. A quantum state may be represented by a vector in Hilbert space or, more generally, by a density matrix ρ . Measurement operators act on this state and determine the probabilities of different outcomes.

The mapping

$$\rho \rightarrow p_k \quad (28)$$

transforms the quantum state into a probability distribution over measurement outcomes. This transformation is linear in the state variables when expressed in terms of expectation values. Consequently, quantum measurement can be viewed as a probabilistic map from a state space to an outcome space.

This viewpoint emphasizes that measurement devices function as probabilistic generators whose output distributions depend on the system state. Such generators can be represented computationally by structures that map state representations to probability distributions.

Neural networks provide one possible computational realization of such mappings.

D. The Born Perceptron Architecture

The Born perceptron is a neural architecture designed to mimic structural aspects of quantum measurement. The outputs of the perceptron are interpreted as value-index pairs

$$(v_1, i_1), (v_2, i_2), \dots, (v_n, i_n). \quad (29)$$

The activations v_k are converted into probabilities

$$p(i_k | x) = \frac{e^{v_k}}{\sum_j e^{v_j}}. \quad (30)$$

Sampling from this distribution produces an index i_k that represents the outcome of a measurement-like process.

In contrast to conventional classification tasks, the indices are not restricted to discrete labels. Instead they may represent coordinates within a geometric manifold

$$i_k \in \mathbb{R}^d. \quad (31)$$

Each index therefore corresponds to a candidate point in a continuous state space. Sampling from the perceptron output selects one of these points.

For example, if the indices correspond to points on the Bloch sphere, the perceptron generates a probability distribution over possible qubit states. Sampling from the distribution produces a specific state vector that may be propagated to subsequent layers.

This interpretation transforms the perceptron into a mechanism that maps an input state representation to a distribution over candidate physical states.

E. Probability Amplitudes and the Born Rule

Quantum measurement probabilities are determined by probability amplitudes. If a quantum state is expanded in a basis with amplitudes ψ_k , the probability of observing outcome k is given by the Born rule

$$p_k = |\psi_k|^2. \quad (32)$$

The use of complex amplitudes allows interference between different contributions to the state. This interference is responsible for many of the distinctive features of quantum phenomena.

A computational architecture that aims to reproduce the structure of quantum probabilities must therefore accommodate complex-valued quantities whose squared magnitudes determine probabilities.

F. Complex Activations and Sequential Sampling

The Born perceptron incorporates this feature by allowing activations to take complex values

$$z_k = w_k \cdot x. \quad (33)$$

Probabilities are then determined by

$$p_k = \frac{|z_k|^2}{\sum_j |z_j|^2}. \quad (34)$$

The complex activations z_k play a role analogous to probability amplitudes, while the normalized squared magnitudes reproduce the Born probability rule.

When multiple Born perceptron layers are arranged sequentially, the sampled index from one layer becomes the input to the next. If the system state at step t is denoted by x_t , the next state is sampled according to

$$x_{t+1} \sim p(i|x_t). \quad (35)$$

This process generates a stochastic trajectory

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \quad (36)$$

The probability of a particular trajectory is

$$P(\text{path}) = \prod_t P(x_{t+1}|x_t). \quad (37)$$

Such trajectories resemble the sequences of measurement outcomes produced when a quantum system is repeatedly observed.

G. Distributed Measurements in Time

Quantum measurements are often distributed in time rather than occurring as isolated events. In such situations the joint probability of a sequence of measurement outcomes depends on the entire trajectory of the system.

The sequential sampling mechanism of stacked Born perceptrons naturally produces such trajectories. Each layer generates a distribution conditioned on the current state and samples an outcome that influences subsequent evolution.

Quantum System

state ρ
 $\downarrow$ measurement operators
probabilities p_k

$\Longleftrightarrow$

Neural Representation

state vector x
 $\downarrow$ perceptron layer
probabilities $p(i|x)$

$\Rightarrow$

Sampled Trajectory

$i_1, i_2, i_3, \dots$

Fig. 1. Conceptual correspondence between quantum measurement and the Born perceptron architecture. A quantum state generates measurement probabilities through operators acting on the state. The Born perceptron produces analogous probability distributions from a neural state vector, and sampling these probabilities generates stochastic trajectories of outcomes.

This architecture therefore provides a computational framework capable of generating stochastic measurement sequences similar to those appearing in quantum trajectory descriptions.

In the following section we extend this framework to incorporate the disturbance associated with measurement. This leads to a neural interpretation of measurement back-action and the quantum nondemolition measurement theory developed by Braginsky and collaborators.

XIII. MEASUREMENT DISTURBANCE AND QUANTUM NONDEMOLITION OBSERVABLES

A. Measurement Disturbance in Sequential Neural Sampling

In the previous section we introduced the Born perceptron as a neural architecture capable of generating stochastic trajectories through sequential sampling of probability distributions. Each layer produces an outcome that becomes the input to the next stage of the computation. This sequential sampling process naturally raises the question of how the act of sampling influences subsequent evolution.

Consider a neural state vector x_t representing the state of the system at step t . A measurement-like layer produces a probability distribution over possible outcomes

$$p(i|x_t) \quad (38)$$

and sampling from this distribution yields an index

$$i_t \sim p(i|x_t). \quad (39)$$

The sampled outcome modifies the input to the next stage, generating a new state x_{t+1} . The evolution of the system therefore depends explicitly on the measurement result. In its most general form the update rule can be written as

$$x_{t+1} = F(x_t, i_t), \quad (40)$$

where F represents the transformation implemented by the network.

This dependence of future states on measurement outcomes implies that the act of sampling introduces a disturbance into the system dynamics. The disturbance arises because the sampling process selects a specific outcome from a distribution that encodes multiple possibilities. Once a particular outcome has been chosen, the subsequent trajectory is constrained by that selection.

From this perspective the Born perceptron does not merely read out a state representation; it modifies the state through the act of measurement. This behavior mirrors a central feature of quantum mechanics, where measurements influence the system being observed.

B. Latent State Dynamics and Disturbance Channels

To analyze the structure of this disturbance it is useful to introduce an explicit representation of the latent state of the system. Let the neural state be represented by a vector

$$s_t = (x_t, p_t), \quad (41)$$

where x_t and p_t may be interpreted as conjugate variables analogous to position and momentum in classical mechanics. Although these variables arise within a neural representation, they provide a convenient way to analyze the effects of measurement disturbance.

Suppose that a measurement layer samples a position-like variable from a distribution determined by the state s_t . The sampling process introduces fluctuations into the conjugate variable p_t . The evolution of the state can then be written as

$$s_{t+1} = H(s_t) + \eta_t, \quad (42)$$

where H represents the deterministic dynamics and η_t is a disturbance term associated with measurement.

The disturbance term may be interpreted as a stochastic kick applied to the latent variables. In particular, sampling of the position coordinate introduces uncertainty into the momentum coordinate. The magnitude of the disturbance depends on the precision with which the measurement is performed. Highly precise measurements correspond to narrow probability distributions and therefore produce larger fluctuations in conjugate variables.

This trade-off between measurement precision and disturbance is a central feature of quantum measurement theory.

C. Connection to Quantum Measurement Back-Action

The disturbance mechanism described above closely resembles the phenomenon of measurement back-action in quantum mechanics. According to the Heisenberg uncertainty principle, position and momentum operators satisfy the commutation relation

$$[x, p] = i\hbar. \quad (43)$$

This relation implies that the uncertainties in position and momentum obey

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (44)$$

As a result, any attempt to measure the position of a particle with high precision inevitably disturbs its momentum. The disturbance subsequently influences the particle's future motion.

The concept of measurement back-action plays a crucial role in precision measurement experiments. In particular, the detection of extremely small displacements, such as those produced by gravitational waves, requires repeated measurements of mechanical oscillators. In such experiments the measurement process itself can introduce noise that limits the achievable sensitivity. This limitation is known as the standard quantum limit.

The neural disturbance mechanism introduced above reproduces the qualitative structure of measurement back-action. Sampling of a position-like variable generates fluctuations in the conjugate variable, which then propagate through subsequent evolution. The resulting trajectories exhibit a stochastic character similar to those observed in quantum measurement processes.

D. Emergence of Quantum Nondemolition Observables

Although many observables are affected by measurement back-action, certain special observables can be monitored repeatedly without being disturbed. Such quantities are known as quantum nondemolition (QND) observables. The theory of QND measurements was developed extensively by Braginsky and collaborators in the context of gravitational-wave detection.

An observable A is said to be a QND observable if measurements performed at different times commute,

$$[A(t_i), A(t_j)] = 0. \quad (45)$$

When this condition is satisfied, a measurement of A does not disturb future measurements of the same observable. The observable can therefore be monitored continuously without being destroyed by the measurement process.

A well-known example occurs for a harmonic oscillator. If $x(t)$ and $p(t)$ denote the position and momentum of the oscillator, one may define the quadrature variables

$$X_1 = x \cos \omega t - \frac{p}{m\omega} \sin \omega t \quad (46)$$

and

$$X_2 = x \sin \omega t + \frac{p}{m\omega} \cos \omega t. \quad (47)$$

These variables correspond to the slowly varying amplitudes of the oscillator motion. Measurements of a single quadrature can be arranged so that the disturbance is transferred entirely to the conjugate quadrature. As a result, repeated measurements

of the chosen quadrature do not degrade its future predictability.

This principle underlies a variety of back-action–evading measurement schemes.

E. Neural Invariants and Back-Action Evading Variables

The concept of QND observables has a natural interpretation within the neural framework. Consider a function of the latent state

$$A_t = g(s_t). \quad (48)$$

If the disturbance term η_t affects only components orthogonal to the gradient of g , then the value of A_t remains unchanged by the measurement process. In this case the variable A_t behaves as an invariant of the neural dynamics.

Mathematically, the condition for invariance may be written as

$$A_{t+1} = A_t. \quad (49)$$

Such invariant features correspond to directions in state space that are unaffected by measurement disturbance. Measurements may strongly perturb other components of the state vector while leaving the invariant variable unchanged.

This structure closely mirrors the behavior of QND observables in quantum mechanics. The neural invariant plays the role of a measurable quantity that can be monitored repeatedly without accumulating disturbance.

F. Stroboscopic Measurement Interpretation

In many physical systems it is not possible to construct a continuously measurable QND observable. However, it is sometimes possible to perform measurements at specific times such that the disturbance introduced by earlier measurements does not influence subsequent observations. This technique is known as stroboscopic measurement.

For a harmonic oscillator, measurements performed at intervals equal to half the oscillation period correspond to times when the commutator of the position operator with itself vanishes,

$$[x(t), x(t + \tau)] = 0 \quad (50)$$

for $\tau = \pi/\omega$.

Under these conditions a sequence of measurements can track the oscillator position with minimal disturbance.

The neural sampling framework provides a natural interpretation of this phenomenon. If measurement layers are activated only at specific steps in the trajectory, the disturbance introduced at one step may evolve into a direction that does not affect the measured variable at the next step. By synchronizing the measurement intervals with the dynamics of the system, it becomes possible to observe certain features repeatedly without degrading their predictability.

This interpretation illustrates how the Born perceptron architecture can reproduce several qualitative aspects of quantum measurement theory, including measurement disturbance, back-action, and the emergence of QND observables.

In the following section we extend this framework to the continuous-time limit and show how sequential neural measurements lead naturally to stochastic evolution equations closely related to the Belavkin theory of quantum filtering.

XIV. CONTINUOUS MEASUREMENT AND NEURAL STOCHASTIC DYNAMICS

A. Continuous Quantum Measurement

Many physical measurement processes occur not as isolated discrete events but as continuous interactions between a system and a measuring device. In such situations information about the system is extracted gradually, producing a measurement record that evolves in time. Examples include optical homodyne detection, superconducting qubit monitoring, and the observation of mechanical oscillators in gravitational-wave detectors.

The theoretical description of continuously monitored quantum systems differs from the instantaneous projection picture of traditional measurement theory. Instead of a single collapse event, the system state evolves stochastically in response to the continuous stream of measurement information. The state conditioned on the measurement record obeys a stochastic differential equation.

Let ρ_t denote the density matrix of a system conditioned on the measurement outcomes up to time t . In the theory developed by Belavkin, the evolution of ρ_t can be written in the form

$$d\rho_t = -i[H, \rho_t]dt + \mathcal{D}[L]\rho_t dt + \mathcal{H}[L]\rho_t dW_t, \quad (51)$$

where H is the Hamiltonian of the system, L is the measurement operator, and dW_t is a Wiener noise increment. The superoperators $\mathcal{D}$ and $\mathcal{H}$ describe decoherence and stochastic state updates induced by the measurement process.

The measurement record itself is described by

$$dY_t = \text{Tr}[(L + L^\dagger)\rho_t]dt + dW_t. \quad (52)$$

The stochastic noise term reflects the unavoidable randomness associated with quantum measurement.

B. Discrete Neural Measurement Updates

The Born perceptron architecture introduced earlier generates measurement outcomes through sequential probabilistic sampling. At step t the system state is represented by a neural vector x_t . A measurement layer computes probabilities

$$p(i|x_t) \quad (53)$$

and produces a sampled outcome

$$i_t \sim p(i|x_t). \quad (54)$$

The sampled index is then used to update the system representation, producing the next state

$$x_{t+1} = F(x_t, i_t). \quad (55)$$

This update rule resembles the filtering step used in sequential inference algorithms. The neural system maintains an internal state that is modified whenever a new observation becomes available. In this sense the perceptron layer acts as a measurement device whose outcomes influence the subsequent trajectory of the latent state.

The stochastic nature of the update rule implies that the evolution of x_t forms a random process. The probability of observing a particular trajectory is determined by the product of conditional probabilities generated at each step.

C. Continuous-Time Limit

To establish a connection with continuous measurement theory we consider the limit in which the time interval between successive measurement steps becomes very small. Let the interval between updates be denoted by Δt . For sufficiently small Δt , the measurement outcome can be decomposed into a deterministic component and a noise component,

$$m_t = \mu(x_t)\Delta t + \sqrt{\Delta t}\eta_t, \quad (56)$$

where η_t is a zero-mean random variable with unit variance.

In the limit $\Delta t \rightarrow 0$, the cumulative effect of the noise terms converges to a Wiener process. The sequence of measurement outcomes therefore approaches a continuous measurement record

$$dY_t = \mu(x_t)dt + dW_t. \quad (57)$$

This expression has the same structure as the measurement signal appearing in the theory of continuous quantum measurement.

The state update rule can also be expanded in powers of Δt . Writing

$$x_{t+\Delta t} = x_t + f(x_t)\Delta t + g(x_t)\eta_t\sqrt{\Delta t}, \quad (58)$$

and taking the limit $\Delta t \rightarrow 0$ yields the stochastic differential equation

$$dx_t = f(x_t)dt + g(x_t)dW_t. \quad (59)$$

The neural state therefore evolves according to a diffusion process driven by measurement noise.

D. Neural Stochastic Dynamics

The stochastic differential equation obtained above describes the evolution of the neural state under repeated measurement updates. The deterministic term $f(x_t)$ represents the intrinsic dynamics of the system, while the stochastic term $g(x_t)dW_t$ captures the fluctuations introduced by measurement.

Such equations are commonly encountered in models of physical systems subject to noise. In the present context

the noise arises from the sampling process used to generate measurement outcomes.

If the neural state is interpreted as a representation of a physical system, the stochastic term can be viewed as the neural analogue of measurement back-action. Each measurement introduces a random perturbation that modifies the state trajectory.

The resulting dynamics produce a family of possible trajectories, each corresponding to a different realization of the measurement noise. The ensemble of trajectories represents the range of possible evolutions compatible with the measurement record.

E. Belavkin Filtering Equation

The stochastic evolution derived above closely resembles the structure of the Belavkin filtering equation used in quantum measurement theory. In that theory the conditional density matrix evolves according to

$$d\rho_t = -i[H, \rho_t]dt + L\rho_t L^\dagger dt - \frac{1}{2}\{L^\dagger L, \rho_t\}dt + \mathcal{H}[L]\rho_t dW_t. \quad (60)$$

The deterministic terms describe the Hamiltonian evolution and measurement-induced decoherence, while the stochastic term represents the random update produced by the measurement record.

The measurement signal itself is

$$dY_t = \text{Tr}[(L + L^\dagger)\rho_t]dt + dW_t. \quad (61)$$

These equations form the basis of quantum filtering theory, which describes how an observer updates their knowledge of a system based on a continuous stream of measurement outcomes.

F. Mapping Neural Dynamics to Quantum Filtering

The correspondence between the neural dynamics and quantum filtering can be summarized as follows.

Quantum Measurement Theory	Neural Representation
Density matrix ρ_t	Neural state vector x_t
Measurement record dY_t	Sampled perceptron output
Wiener increment dW_t	Sampling noise
Measurement operator L	Perceptron weight matrix
Conditional state update	Neural state update

Under this interpretation the Born perceptron architecture functions as a computational mechanism that generates stochastic trajectories analogous to those arising in continuously monitored quantum systems.

The discrete sampling steps of the perceptron correspond to measurement events, while the continuous-time limit reproduces the stochastic evolution characteristic of quantum filtering equations.

In the following section we examine the statistical interpretation of these stochastic dynamics and connect the neural measurement framework with the statistical analysis of

quantum trajectories developed by Belavkin and later extended by Guță.

XV. STATISTICAL FILTERING AND QUANTUM ENERGY TELEPORTATION

A. Measurement Records and Quantum Trajectories

Continuous quantum measurement generates a time-dependent record of outcomes that reflect the interaction between a physical system and its measurement apparatus. In quantum trajectory theory the system state evolves conditionally on this measurement record. Each realization of the measurement signal corresponds to a stochastic trajectory of the system state.

Let $Y(t)$ denote the measurement record produced by monitoring a quantum observable. In the framework of continuous quantum measurement the signal may be written as

$$dY_t = \text{Tr}[(L + L^\dagger)\rho_t]dt + dW_t, \quad (62)$$

where ρ_t is the conditional density matrix of the system and dW_t is a Wiener noise increment. The first term represents the expected signal determined by the current system state, while the stochastic term represents unavoidable measurement noise.

The conditional state ρ_t evolves according to a stochastic master equation, often referred to as the Belavkin filtering equation. The measurement record therefore provides a continuous stream of information that allows an observer to update their estimate of the system state.

Each realization of the measurement noise produces a different trajectory of the conditional state. The ensemble of trajectories describes the possible evolutions of the system under continuous observation.

B. Neural Representation of Measurement Records

Within the Born perceptron framework the measurement record appears naturally as the sequence of outcomes generated by the sampling process. At each time step the neural architecture produces an outcome

$$i_t \sim p(i|x_t) \quad (63)$$

which becomes part of a trajectory

$$i_1, i_2, i_3, \dots \quad (64)$$

This sequence is directly analogous to the measurement record produced in a physical experiment. The internal neural state x_t evolves in response to these observations according to the update rule

$$x_{t+1} = F(x_t, i_t). \quad (65)$$

From a statistical perspective the neural system maintains a latent representation of the system state that is refined whenever new measurement outcomes become available. The perceptron layers therefore implement a filtering process in which the internal state acts as an evolving estimate of the underlying physical system.

The stochastic trajectories generated by the neural architecture correspond to realizations of a measurement process in which both the state estimate and the measurement record evolve simultaneously.

C. Belavkin Filtering as Bayesian Inference

The theory of quantum filtering developed by Belavkin provides a mathematical framework for describing the conditional evolution of quantum systems under continuous observation. In this theory the density matrix ρ_t represents the observer's best estimate of the system state given the measurement record up to time t .

The update of ρ_t can be interpreted as a quantum analogue of Bayesian inference. Each new measurement outcome modifies the observer's knowledge of the system and leads to a revised estimate of the system state.

This interpretation emphasizes that quantum measurement theory is fundamentally concerned with the processing of information obtained from measurement records. The stochastic master equation governing ρ_t describes how information extracted from the system modifies the observer's description of the system.

In this sense the Belavkin equation plays a role analogous to classical filtering equations used in statistical signal processing.

D. Neural Quantum Filters

The Born perceptron architecture provides a computational mechanism capable of implementing a similar filtering process. The neural state vector x_t represents the internal estimate of the system state, while the sampled outcomes i_t correspond to measurement observations.

The update rule

$$x_{t+1} = F(x_t, i_t) \quad (66)$$

updates the internal state based on new observations. This structure closely resembles the update steps used in classical filtering algorithms such as the Kalman filter or particle filter.

The perceptron weights determine the measurement model that maps internal states to observable outcomes. Learning these weights corresponds to identifying the measurement operators that generate the observed data. Consequently the neural architecture may be interpreted as a learnable quantum filter capable of modeling measurement processes.

This perspective suggests a natural synergy between neural computation and the statistical analysis of quantum measurement records.

E. Belavkin–Guță Statistical Interpretation

The statistical properties of quantum filtering processes have been studied extensively by Guță and collaborators. Their work analyzes the asymptotic behavior of measurement records produced by continuously observed quantum systems and investigates the problem of estimating unknown system parameters.

In this framework the measurement record is treated as a stochastic process whose statistical properties encode information about the underlying system dynamics. By analyzing the measurement data one can infer parameters such as coupling strengths or Hamiltonian coefficients.

The Born perceptron architecture generates stochastic measurement records through its sampling process. These records can therefore be analyzed using statistical techniques similar to those employed in quantum filtering theory.

The combination of neural generative models and quantum statistical inference provides a promising framework for analyzing measurement processes in complex quantum systems.

F. Quantum Energy Teleportation

An important application of measurement back-action arises in the phenomenon known as quantum energy teleportation (QET), introduced by Hotta. In QET protocols energy can effectively be transported between spatially separated regions of a quantum system using only local operations and classical communication.

The key idea underlying QET is that local measurements can inject energy into a quantum field or many-body system through measurement back-action. Because quantum systems exhibit spatial correlations in their ground states, information obtained from a local measurement can be used to perform operations in a distant region that extract energy from the system.

In a typical QET protocol two observers, traditionally referred to as Alice and Bob, interact with a shared quantum system. Alice performs a local measurement on the system at position A . The measurement injects energy into the system due to back-action. Alice then communicates the measurement result to Bob through a classical channel. Bob uses this information to perform a local operation at position B that allows him to extract energy from the system.

Remarkably, the energy extracted by Bob can exceed the energy locally present in the system prior to his operation. The protocol effectively transfers energy between distant locations without any physical energy carrier traveling between them.

This phenomenon does not violate energy conservation because the total energy of the system increases when Alice performs her measurement. The measurement back-action provides the energy that later becomes available to Bob.

G. Back-Action and Energy Injection

The QET protocol highlights the fundamental role of measurement back-action as a physical resource. When a measurement is performed on a quantum system, the interaction between the system and the measuring device can transfer energy to the system.

In conventional measurement theory this effect is often regarded as a source of noise or disturbance. In QET, however, the back-action is deliberately exploited to inject energy into the system in a controlled manner.

The energy injection can be analyzed by considering the expectation value of the system Hamiltonian before and after

the measurement. Let H denote the Hamiltonian of the system. If ρ is the initial state and ρ' is the post-measurement state, the energy change produced by the measurement is

$$\Delta E = \text{Tr}(H\rho') - \text{Tr}(H\rho). \quad (67)$$

In many QET protocols the measurement produces a positive energy injection that can later be partially extracted elsewhere in the system.

H. Neural Representation of Quantum Energy Teleportation

The Born perceptron framework provides a natural computational model for representing the measurement processes that underlie QET protocols. In this representation the neural state vector x_t corresponds to the quantum state of the system, while perceptron layers represent measurement operations.

Alice's measurement corresponds to sampling an outcome

$$i_A \sim p(i|x_t) \quad (68)$$

which modifies the neural state according to

$$x_{t+1} = F(x_t, i_A). \quad (69)$$

This update represents the back-action of the measurement, which injects energy into the system.

The measurement outcome is then transmitted through a classical communication channel and used to condition Bob's operation. In the neural model this corresponds to applying a transformation

$$x_{t+2} = G(x_{t+1}, i_A) \quad (70)$$

that depends explicitly on Alice's measurement result.

Bob's operation extracts energy from the system by exploiting correlations present in the system state. The effectiveness of the extraction depends on the information contained in the measurement outcome.

I. Trajectory Interpretation of QET

Within the neural framework the QET protocol can be interpreted as a trajectory in which measurement outcomes guide subsequent operations. The trajectory consists of three stages:

- 1) Measurement at location A producing outcome i_A .
- 2) State update due to measurement back-action.
- 3) Conditional operation at location B using the communicated outcome.

Each stage modifies the neural state vector and contributes to the overall trajectory of the system.

The probability of a particular sequence of events is given by

$$P(\text{path}) = P(i_A|x_t)P(x_{t+1}|x_t, i_A)P(x_{t+2}|x_{t+1}, i_A). \quad (71)$$

This expression highlights the sequential nature of the protocol and the dependence of later operations on earlier measurement outcomes.

J. Implications for Quantum Measurement Models

The QET protocol provides a compelling example of how measurement back-action can be harnessed as a useful physical resource. Rather than treating measurement disturbance purely as a limitation, QET demonstrates that the energy injected by measurement can be used to perform useful tasks.

Within the Born perceptron framework, this phenomenon appears as a consequence of the state update induced by the sampling process. The measurement outcome modifies the neural state in a way that alters the system's energy distribution.

Because the neural architecture explicitly represents measurement outcomes and state updates, it provides a natural computational framework for modeling protocols that exploit measurement-induced energy transfer.

K. Summary

The framework developed in this work connects several important ideas in quantum measurement theory and neural computation. The Born perceptron architecture provides a neural representation of measurement processes in which probabilistic sampling generates trajectories of outcomes. Measurement back-action appears as stochastic disturbances in the neural state, reproducing key features of quantum measurement theory.

In the continuous-time limit the sequential sampling process leads to stochastic differential equations analogous to the Belavkin filtering equation. The measurement outcomes form a stochastic record that can be analyzed using the statistical methods developed by Belavkin and Guță.

The example of quantum energy teleportation illustrates how measurement back-action can serve as a physical resource for energy transfer in correlated quantum systems. Within the neural framework this process corresponds to state updates and conditional operations guided by measurement outcomes.

These connections suggest that neural architectures based on probabilistic sampling may provide useful computational tools for modeling complex quantum measurement processes and exploring new applications of measurement-induced dynamics.

ACKNOWLEDGMENTS

This work was produced with the assistance of large language models including ChatGPT 5.2, Google Gemini and Grok.

REFERENCES

- [1] M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.
- [2] C. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [3] R. Feynman and A. Hibbs, *Quantum Mechanics and Path Integrals*. McGraw-Hill, 1965.
- [4] C. M. Caves, "Quantum mechanics of measurements distributed in time: A path-integral formulation," *Physical Review D*, vol. 33, no. 6, 1986.